La Scuola d'Italia / Huson / Physics
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Nickname: SOLUTIONS

1.9 Rate Problems

Reference (provided). rate × time = quantity (for motion: $r \cdot t = d$)
1 km = 1000 m 1 m = 100 cm 1 h = 3600 s 1 min = 60 s 1 gal ≈ 3.785 L

Solve each problem in two steps. First use rate × time = quantity, or a rearrangement of it. Then convert your result to the units the question asks for. Round your final answer to the number of significant figures the data supports.

1. The Sun is 1.496×10^8 km from the Earth. Light travels at 3.00×10^8 m/s. How many minutes does sunlight take to reach the Earth?

$d = 1.496 \times 10^8$ km

$r = 3.00 \times 10^8$ m/s $= 3.00 \times 10^5$ km/s

$d = rt$, $t = \frac{d}{r}$

$t = \frac{1.496 \times 10^8 \text{ km}}{3.00 \times 10^5 \text{ km/s}} = 498.666\ldots$ s

$t = 498.\bar{6}$ s $\times \frac{1 \text{ min}}{60 \text{ sec}} = 8.3\bar{1}$ m

≈ 8.31 minutes

2. Hair grows about 15 centimeters per year. A student does not cut their hair from age 9 to age 15. By how many meters does it grow?

$t = 15 - 9 = 6$ years $d = rt$

$d = 6\text{y} \times \frac{15 \text{ cm}}{\text{y}} = 90$ cm

$d = 90 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ c}} = 0.90$ m

Quantity = Rate · Time

3. A US kitchen faucet lets through at most 2.2 gallons of water per minute. Filling a large pasta pot takes 45 seconds. How many liters of water go into the pot?

$\text{Amount} = 2.2 \frac{\text{gal}}{\text{min}} \times 0.75 \text{ min}$ amount = r · t

45 seconds = 0.75 min

$= 1.65$ gallon

$\text{Amount} = 1.65 \text{ g} \times \frac{3.785 \text{ L}}{1 \text{ g}}$

$= 6.24525\ldots \text{ L}$

$\approx 6.2 \text{ L}$

4. Usain Bolt's 100-meter world record is 9.58 seconds. What was his average speed in kilometers per hour?

$r = \frac{100 \text{ m}}{9.58 \text{ s}} = 10.438\ldots$ m/s

$r = 10.438\ldots \frac{\text{m}}{\text{s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ hr}} = 37.578\ldots \frac{\text{km}}{\text{hr}}$

$= 37.6 \frac{\text{km}}{\text{hr}}$